



HACKOWATT 2026

AI for Green Energy:
Smart Energy Challenge

ALL SCENARIOS

Common Challenge Assumptions

The following assumptions apply to all scenarios.

01

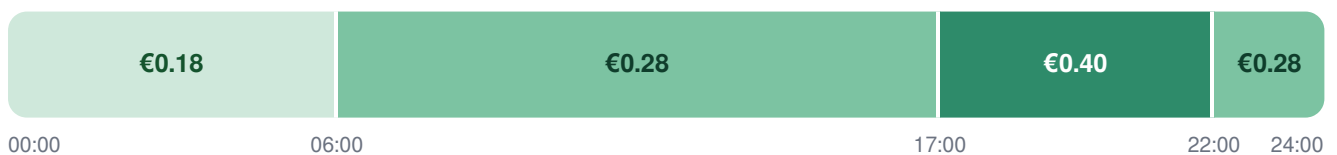
Minimum requirements

- Use a **minimum of 30 consecutive days** of historical data with hourly resolution.
- Use **real historical and forecast weather data** for the location specified in your scenario. Example source: **Open-Meteo**. Other reliable sources may also be used.
- At minimum, include **outdoor temperature** as a weather-related variable.
- Clearly document any additional assumptions introduced in your solution.

02

Electricity tariff

For all scenarios, use the following simplified tariff:



TIME

ELECTRICITY PRICE

00:00–06:00	€0.18/kWh
06:00–17:00	€0.28/kWh
17:00–22:00	€0.40/kWh
22:00–00:00	€0.28/kWh

03

Renewable Energy Simulator

For comparison purposes, use the following common assumptions:

€1,300 /kWp

PV installation cost

€0.08 /kWh

Electricity exported
to the grid

1%

Annual PV operating
cost
of the initial investment cost

PV production should reflect the location specified in the assigned scenario.

04

Data sources

Suggested free sources for weather and PV production data:

WEATHER · FORECAST

Open-Meteo

Free, open-source weather
API with hourly forecasts,
including outdoor temperature.

open-meteo.com ↗

WEATHER · HISTORICAL

Open-Meteo Historical Weather API

Hourly historical weather data
for any location, e.g.
temperature_2m.

archive-api.open-meteo.com



PV PRODUCTION

JRC – PVGIS

European Commission tool
estimating solar PV production
for a given location
and installation.

[PVGIS tools · JRC](#) ↗

05

Want to go further?

You are welcome to extend your solution with additional weather variables, longer historical datasets, advanced forecasting methods, battery storage, dynamic tariffs or additional optimisation features.

These extensions are optional — focus first on delivering a clear and working solution to the core challenge.

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